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Modular assembly of complex *C*-alkyl glycoside enabled via cooperative NHC and photoredox catalysis

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C-Alkyl glycosides are found in natural products and are indispensable in drug development programs, owing to their diverse range of medicinal properties. Two-component reactions involving glycosyl radical addition are well documented and proved efficient for the *C*-alkyl glycosides synthesis. However, multicomponent reactions for the rapid and controllable construction of structurally diversified *C*-alkyl glycosides in a single step remains a significant challenge due to the issues associated with balancing reactivity, compatibility, and selectivity. Herein, we report an NHC and photoredox co-catalyzed three-component method to access complex and valuable *C*-alkyl glycosides. This protocol exhibits broad substrate scope and excellent chemo- and regioselectivity. Furthermore, nine types of sugars are compatible with this reaction. In addition, the generality of this catalytic strategy was further highlighted by the potential for scalable production, the feasibility of further modifications of the products and its successful application in the late-stage functionalization of pharmaceutical skeletons.

***C*-alkyl glycosides, three-component, NHC catalysis, visible-light photoredox catalysis, radical-radical coupling**

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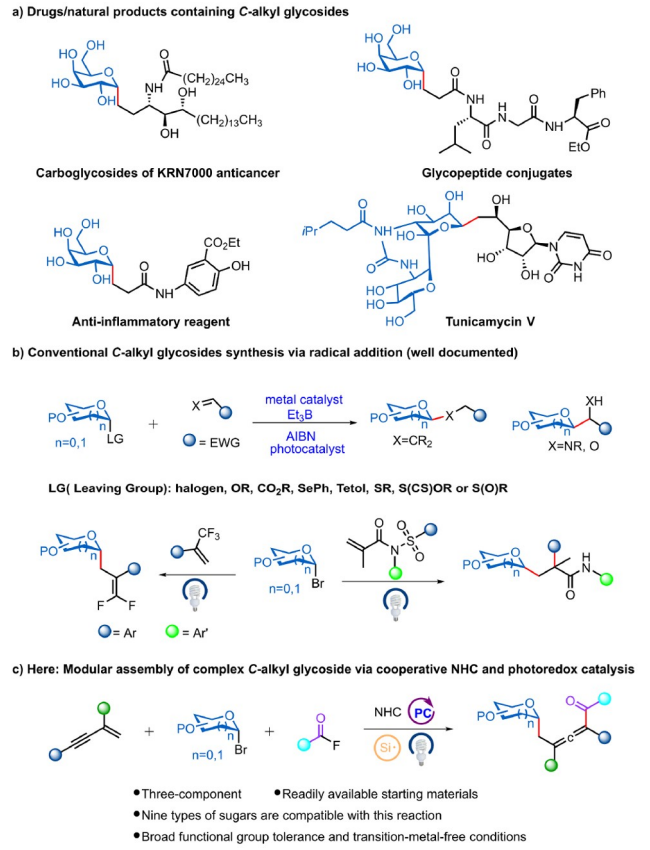
1 Introduction

C-Alkylglycosides are an important class of carbohydrate scaffolds that are prevalent structural motifs in biologically relevant molecules as key structural elements (Scheme 1a) [1]. Compared with *O*-glycosides, their stability *in vitro* and *in vivo* is obviously enhanced, which plays an important role in the modern design of *C*-glycoside drug candidates and studies of important biological processes [2].

Considering the important biological activity and medical value of *C*-alkylglycosides, as such, the development of synthetic methods for *C*-alkylglycosides is an unmet need for organic chemists. The generation of glycosyl radical intermediates provides an efficient strategy for the synthesis of *C*-glycosides [3–10]. Among those approaches, the addition of glycosyl radicals across alkenes has evolved to a key strategy for the synthesis of *C*-alkylglycosides (Scheme 1b) [3,7–10]. Inspired by Giese's pioneering work [9a] on the use of tributyltin hydride under photolysis conditions, so far, many research groups have explored the Sn- [9a,9e], Et_3B- [9m,9n], AIBN- [9h], or Sm-mediated [7a,7c,7d] and Ni- [9f], Ru- [9j,9i], Fe- [9o], or Ti-catalyzed [9l] Giese reactions for *C*-alkylglycosides. These reactions, which can be conducted under practical and mild conditions, feature high functional group tolerance and broad substrate scope. Recently, Hu's group [11] realized the synthesis of gem-difluoroalkene-containing *C*-alkylglycosides via a photomediated defluorinative gem-difluoroallylation reaction. Very recently, our group reported the synthesis *C*-alkyl glycosides of *α*-aryl-*β*-glycosamide via Smiles rearrange-

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Scheme 1 (Color online) (a) Drugs/natural products containing *C*-alkyl glycosides. (b) Conventional *C*-alkyl glycosides synthesis via radical addition. (c) Here: modular assembly of complex *C*-alkyl glycoside via cooperative NHC and photoredox catalysis.

ment [12]. Despite the significant progress recorded in this field, the intermolecular three-component cascade reactions for molecular complexity *C*-alkylglycosides synthesis has been far less developed. There is only one example, in 2023, Ackermann's group [13] achieved a domino *meta*-C–H ethylglycosylation by merging the addition of glycosyl radicals to olefins and ruthenium catalysis for distal C–H activation three-component radical cross-coupling reactions. As a result, the development of a new method involving the glycosyl radical intermolecular three-component cascade reactions for complex molecules *C*-alkylglycosides synthesis is greatly desired.

In recent years, the combination of visible light and NHC cocatalyzed radical coupling reactions has generated recent excitement to drive complex bond constructions. The key step of the reaction is radical and ketyl radical cross-coupling delivers a ketone-containing product. So far, many elegant works have been well-established by Ohmiya and other groups [14]. On the other hand, 1,4-difunctionalization of 1,3-enynes via a radical pathway has also been intensively investigated recently [15]. However, NHC-promoted radical 1,4-difunctionalization of 1,3-enynes remains a challenge for the synthesis of functionalized allenes ketones. Du [16], Huang [17], Hong [18] and Zhang *et al.* [19] independently developed a NHC-promoted radical 1,4-difunctionalization of 1,3-enynes, providing a ready access to different substituted allenyl ketones. Despite significant progress, it is surprising to observe that NHC-promoted radical 1,4-glycosylacylation of 1,3-enynes has not yet been explored. Considering the importance of the complex molecules *C*-alkylglycosides in medicinal chemistry and inspired by previous work, we wondered whether the combination of glycosyl radical addition and NHC and photoredox catalysis could be feasible for modular assembly of complex *C*-alkylglycosides. With our group's ongoing research on *C*-glycosylation, herein, we report an NHC and photoredox cocatalyzed three-component reaction from readily available 1,3-enynes, aroyl fluorides, glycosyl bromides. This transformation involves the coupling of ketyl radical from single electron transfer (SET) reduction of *in situ* formed acylazolium ions and allenyl radical from glycosyl radicals selectively attacks the olefin unit of the 1,3-enyne to produce the propargyl radical intermediate, which could undergo reversible resonance, providing direct access to an array of complex and structurally diversified valuable *C*-alkyl glycosides under mild, transition-metal-free conditions.

2 Experimental

In a 10 mL pre-dried schlenk tube with stir bar, $(TMS)_3SiOH$ (39 mg, 0.15 mmol, 1.5 equiv.), glycosylbromide **2a** (0.25 mmol, 2.5 equiv.), **PC1** (4.2 mg, 3 mol%), **NHC-A** (8.3 mg, 20 mol%), and K_2CO_3 (41.4 mg, 0.3 mmol, 3.0 equiv.) were added. The tube was charged with argon, which repeated more than three times. Subsequently, 1 mL MeCN was injected and **1a** (0.1 mmol) and **3a** (0.2 mmol) were added. Then the reaction mixture was irradiated with blue light-emitting diode (LED) at room temperature for 12 h. After the reaction completed, the solvent was evaporated under vacuum. The residue was purified by flash chromatography on silica gel (petroleum ether/ethyl acetate = 3:1) to afford the corresponding product **4a**.

3 Results and discussion

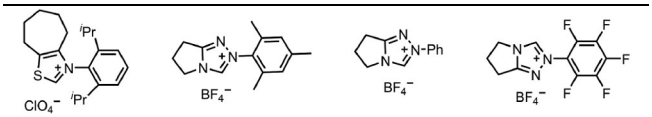
As shown in Table 1, we commenced our investigation by using a 1,3-enyne **1a**, glycosyl bromides (**2a)**, and benzoyl fluoride (**3a**) as the prototype substrates. After extensive exploration of different experimental parameters, we found that upon irradiation of **1a**, **2a** and **3a** in MeCN with blue LEDs at room temperature under argon in the presence of **PC1** as the photocatalyst, $(TMS)_3SiOH$ as the reducing agent, potassium carbonate as bases, and **NHC-A** as the

Table 1 Optimization of reaction conditions[a)]



1a 2a 3a 4a

Entry	Variation from standard conditions	Yield[b)] (%)
1	none	70[c)]
2	**PC2** instead of **PC1**	21
3	**PC3** instead of **PC1**	24
4	**PC4** instead of **PC1**	42
5	$(TMS)_3SiH$ instead of $(TMS)_3SiOH$	40
6	DMF instead of MeCN	0
7	DMSO instead of MeCN	trace
8	K_3PO_4 instead of K_2CO_3	8
9	Na_2CO_3 instead of K_2CO_3	0
10	Cs_2CO_3 instead of K_2CO_3	trace
11	NHC-B instead of NHC-A	12
12	NHC-C instead of NHC-A	10
13	NHC-D instead of NHC-A	0
14	Without $(TMS)_3SiOH$ or **PC1**	0
15	No NHC	0
16	In dark	0
17	air	trace



NHC-A NHC-B NHC-C NHC-D

a) Conditions: **1a** (0.1 mmol, 1.0 equiv.), **2a** (0.25 mmol, 2.5 equiv.), **3a** (0.2 mmol, 2.0 equiv.), **PC1** (3.0 mol%), K_2CO_3 (3.0 equiv.), $(TMS)_3SiOH$ (1.5 equiv.), **NHC-A** (20 mol%) in MeCN (1.0 mL) under blue LEDs at room temperature under argon for 12 h. b) Isolated yield. c) dr = 1.1:1. **PC1** = 4-BrCZIPN, **PC2** = $Ir(ppy)_2(dtbbpy)PF_6$, **PC3**= $Ir[dF(CF_3)ppy_2(dtbbpy)]PF_6$, **PC4** = 4CZIPN.

NHC catalyst, we could obtain target 1,4- addition product **4a** in 70% yield (entry 1). Screening of photosensitizers showed that 4-BrCZIPN exhibited the highest yield (entries 2–4). When the reducing agent was $(TMS)_3SiH$, the yield down to 40% (entry 5). It should be mentioned that the solvent could be the key factor in this catalytic system to control the yield, and no product was observed with dimethyl formamide (DMF), dimethyl sulfoxide (DMSO) (entries 6–7). Replacing K_2CO_3 with other bases led to sharply lower yields (entries 8–10). When other NHC precursors were used, the yield of **4a** decreased substantially (entries 11–13). Control experiments showed that reducing agent $(TMS)_3SiOH$, photosensitizer, NHC catalyst, light and argon were crucial to the reaction (entries 14–17).

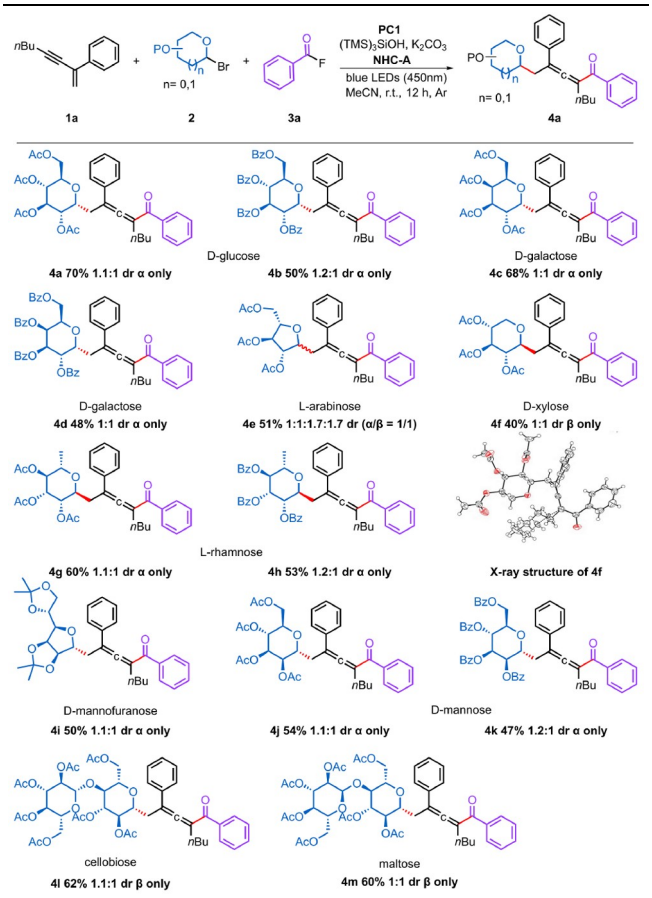
Having established the optimal reaction conditions, and considering the diversity of *C*-glycosides, pyranose and furanose bromide substrates were then investigated. As shown Table 2, both pyranoses and furanoses could give the corresponding allenyl ketone in medium to good yields. Glucopyranoses and galactopyranoses, which protected by different substituents, such as acetoxyl (**4a** and **4c**) and benzoyl (**4b** and **4d**) groups, were well-tolerated to this protocol, affording the desired products in yields ranging from 48% to 70%. The L-arabinose generated product **4e** in 51% yield with the mixed configuration. The reaction of D-xylose also proceeded smoothly to give products **4f** in 40% yield. The protocol could also well accommodate L-rhamnose, and the target products were isolated in 53%–60% yield (**4g**, **4h**). Then we explored D-mannofuranose, delivering the desired products **4i**. D-Mannopyranoses was also examined and gave the corresponding product (**4j, 4k**) in 47%–54% yield. In addition, this reaction also showed good compatibility toward disaccharides (cellubiose and maltose), and the desired product were obtained in 60%–62% yield (**4l**, **4m**).

We subsequently investigated the generality of 1,3-enynes. As shown in Table 3, 1,3-enynes bearing electron-donating (–OMe, –Me) or -withdrawing substituents (–F, Cl, Br) at the *para*-positions of the 2-phenyl rings (**5a**–**5e**) exhibited good reactivity and promoted reactions smoothly in good yields (56%–68%). The substituents at the *ortho*-position or *meta*-position also worked well to provide the target product (**5f**–**5i**) in moderate to good yields (53%–67%). Furthermore, substrates with 1-naphthalene and heterocycles were also amenable to this catalytic process and afforded the desired products (**5**j, **5k**) in comparable yields. Besides, 2-alkyl substituted 1,3-enynes were applicable, giving the desired products (**5l**).

We also explored the scope of acyl fluorides. As shown in Table 4, The electron-donating groups at the aromatic ring's *para*-, *meta*-, or *ortho*-position reacted efficiently to furnish the corresponding products under these reaction conditions in 65%–72% yields (**6a**, **6b**, **6e**, **6g**), while the electron-withdrawing groups at the aromatic ring's *para*- or *ortho*-position led to low efficiency (**6c**, **6d**, **6f**). Furthermore, multi-substituted aryl acyl fluorides were also tolerated and delivered the expected products **6h** in yields of 62%. Substrates with 2-naphthalene moieties could also give the desired products in 51% yield **6i**. Additionally, to our surprise, heterocycles like thiophene, furan and pyridine, which have frequently been found in medicinal chemistry due to their various biological activities, were found to be compatible with the reaction conditions (**6j**–**6l**).

To further demonstrate the potential synthetic application of this protocol, this approach is also applicable to the late-stage drug derivative. Gratifyingly, various drugs/drug fragments linking molecules were successfully functionalized (Scheme 2a), which gave the corresponding products

Table 2 Substrate scope with respect to the glycosyl bromides[a)]

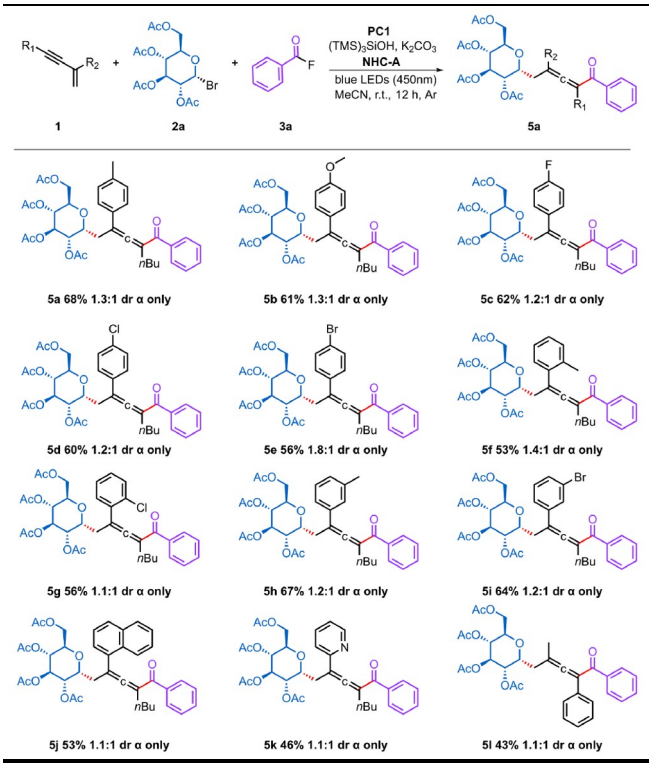


a) Conditions: **1a** (0.1 mmol, 1.0 equiv.), **2** (0.25 mmol, 2.5 equiv.), **3a** (0.2 mmol, 2.0 equiv.), **PC1** (3.0 mol%), K_2CO_3 (3.0 equiv.), $(TMS)_3SiOH$ (1.5 equiv.), **NHC-A** (20 mol%) in MeCN (1.0 mL) under blue LEDs at room temperature under argon for 12 h.

(**6m**–**6p**) in 24%–56% yield. We also conducted scale-up synthesis and further chemical transformations of the products (Scheme 2b). The target product (**4a**) was obtained in 57% yield when the reaction was expanded to the 2.0 mmol level and the ketone unit of **4a** can be reduced by $NaBH_4$. In addition, the bromo-containing product **5e** could undergo palladium-catalyzed Suzuki coupling with 1-cyclopentenylboronic acid to deliver product **6r** (Scheme 2c).

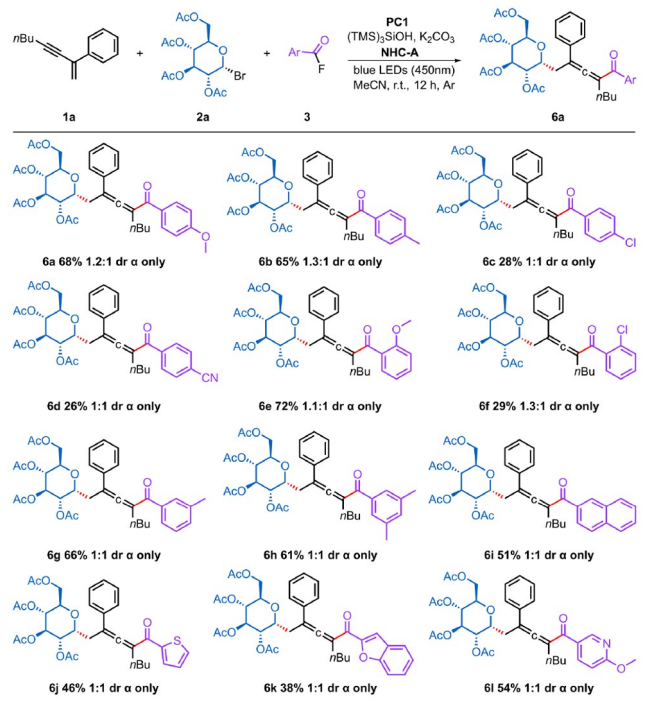
Next, we turned our attention to the mechanism of this transformation. In order to confirm whether radicals were involved, 2,6,6-tetramethyl-1-piperidinyloxy (TEMPO) as a radical scavenger was added under the standard reaction conditions. It was found that the formation of **4a** was completely suppressed, which indicated that a radical mechanism might be involved in this transformation. From this reaction mixture, the adducts of TEMPO with glycosy and ketyl radical were detected by high resolution mass spectrum (HRMS, Figure 1a**)**. When 1,1-diphenylene (3 equiv.) was used to trap the corresponding sugar radical intermediate, the trapped product **6u** was detected by HRMS (Figure 1a), which suggested that this protocol advances via a glycosyl

Table 3 Substrate scope of diverse 1,3-enynes[a)]

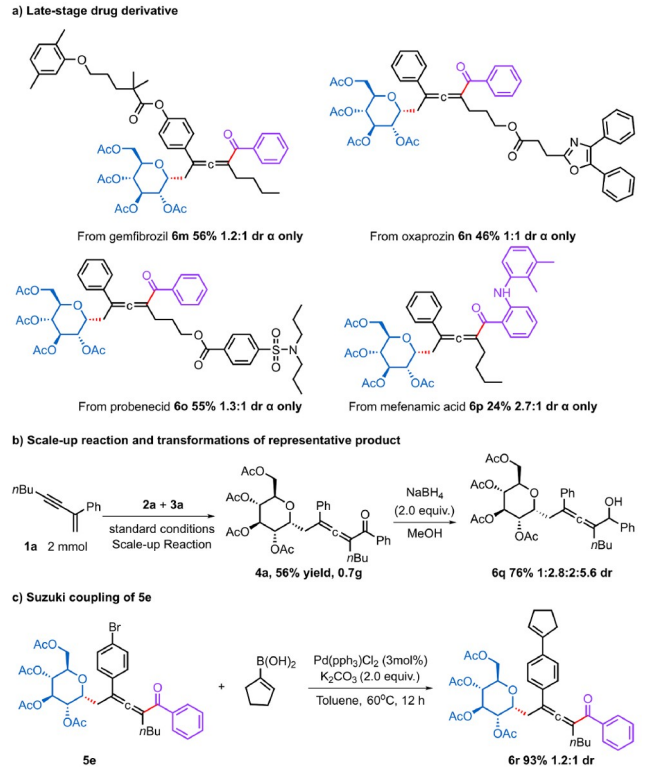


a) Conditions: **1** (0.1 mmol, 1.0 equiv.), **2a** (0.25 mmol, 2.5 equiv.), **3a** (0.2 mmol, 2.0 equiv.), **PC1** (3.0 mol%), K_2CO_3 (3.0 equiv.), $(TMS)_3SiOH$ (1.5 equiv.), **NHC-A** (20 mol%) in MeCN (1.0 mL) under blue LEDs at room temperature under argon for 12 h.

Table 4 Substrate scope with respect to the acyl fluoride[a)]



a) Conditions: **1a** (0.1 mmol, 1.0 equiv.), **2a** (0.25 mmol, 2.5 equiv.), **3** (0.2 mmol, 2.0 equiv.), **PC1** (3.0 mol%), K_2CO_3 (3.0 equiv.), $(TMS)_3SiOH$ (1.5 equiv.), **NHC-A** (20 mol%) in MeCN (1.0 mL) under blue LEDs at room temperature under argon for 12 h.



Scheme 2 (Color online) Late-stage drug derivative, scale-up synthesis and transformations of representative product.

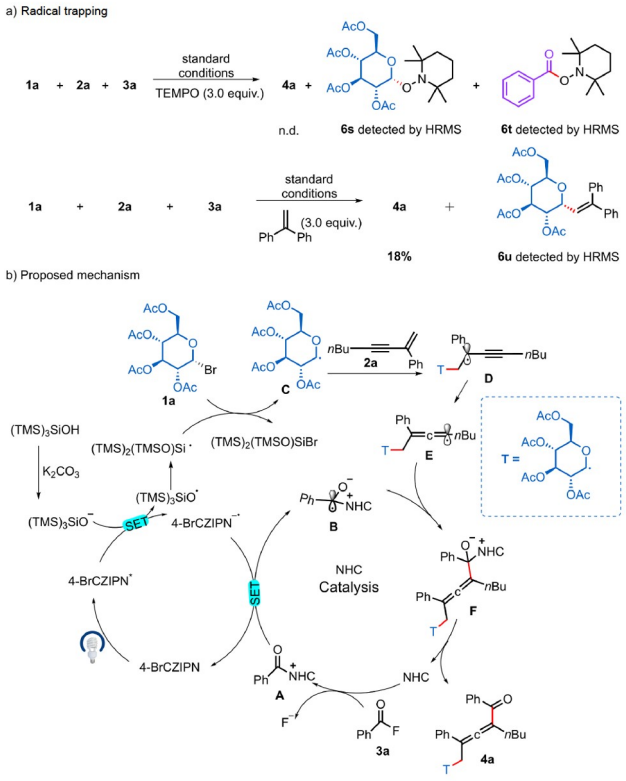


Figure 1 (Color online) (a) Radical trapping. (b) Proposed mechanism.

radical pathway. On the basis of the above results and literature reports, a plausible reaction mechanism for the modular assembly of complex *C*-alkyl glycoside is proposed in (Figure 1b). Initially, the $(TMS)_3SiO^-$ anion, which comes from $(TMS)_3SiOH$ in the presence of K_2CO_3, undergoes single-electron oxidation by the excited 4-BrCZIPN* to afford the $(TMS)_3SiO^{\cdot}$ radical. Subsequently, the $(TMS)_3SiO^{\cdot}$ radical spontaneously turns into the $(TMS)_2(TMSO)Si^{\cdot}$ radical by Brook rearrangement. Next, glycosyl radical **C** is generated from radical transfer between $(TMS)_2(TMSO)Si^{\cdot}$ and glycosyl bromide **1a**, and it selectively attacks the olefin unit of the 1,3-enyne **2a** to produce the propargyl radical intermediate **D**, which could undergo reversible resonance to generate relatively stable allenyl radical **E**. The acyl fluoride **3a** could react with NHCs produces the acylazolium intermediate **A**, which can be reduced by 4-BrCZIPN$^{-\cdot}$ species to provide ketyl radical **B** and regenerate the ground state 4-BrCZIPN photocatalyst. The coupling of **E** with radical **B** ultimately yields the target product **4a** and releases the NHCs catalyst.

4 Conclusions

In summary, we have developed a strategy to accomplish multicomponent assembly of complex *C*-alkyl glycoside via cooperative NHC and photoredox catalysis under mild, transition-metal-free conditions. This multicomponent coupling reaction used readily available starting materials with broad functional group compatibility and excellent chemo- and regionselectivity. Mechanistic studies suggest that this transformation proceeds via radical/radical cross coupling of ketyl radicals with allenyl radicals. The ketyl radicals are generated via SET reduction of *in situ* formed acylazolium ions whereas the allenyl radicals derive from chemo-specific glycosyl radical addition to the 1,3-enynes. We expect that this protocol will provide a versatile platform for the multicomponent assembly of complex *C*-alkyl glycosides and believe this work will stimulate more research in multicomponent reactions for the preparation of structurally diversified valuable *C*-alkyl glycosides that are difficult-to-made in future.

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Conflict of interest The authors declare no conflict of interest.

Supporting information The supporting information is available online at http://chem.scichina.com and http://link.springer.com/journal/11426. The supporting materials are published as submitted, without typesetting or editing. The responsibility for scientific accuracy and content remains entirely with the authors.

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